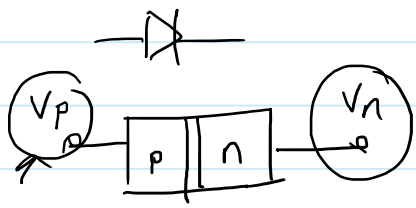
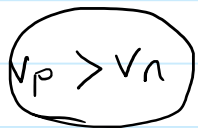
Reading & Discussion

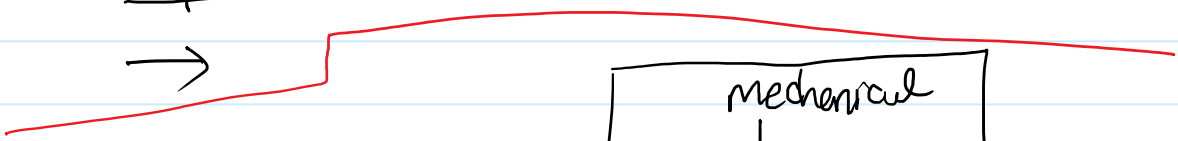
Adel S. Sedra and Kenneth C. Smith, **Microelectronic Circuits** 7th Edition, *Oxford University Press*, 2014.

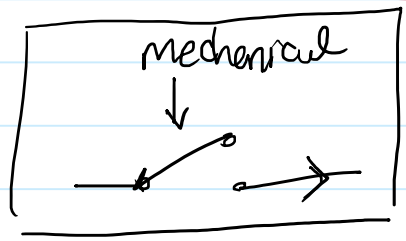
- Chapter 6.1

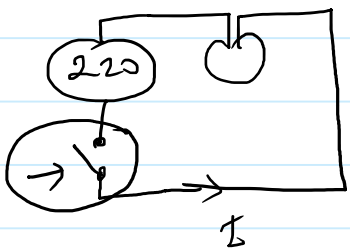


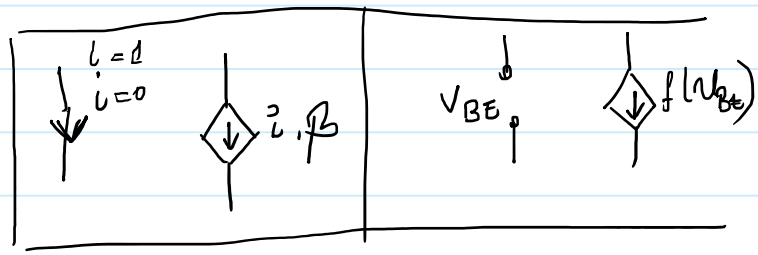


$V_p - V_n \geq 0.7$





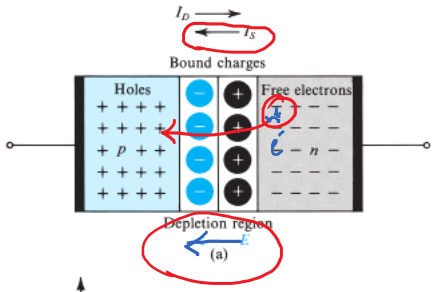


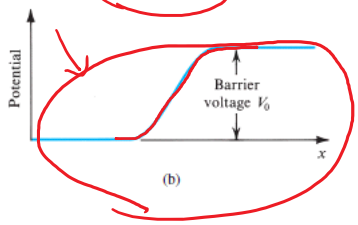




BJT

Recap





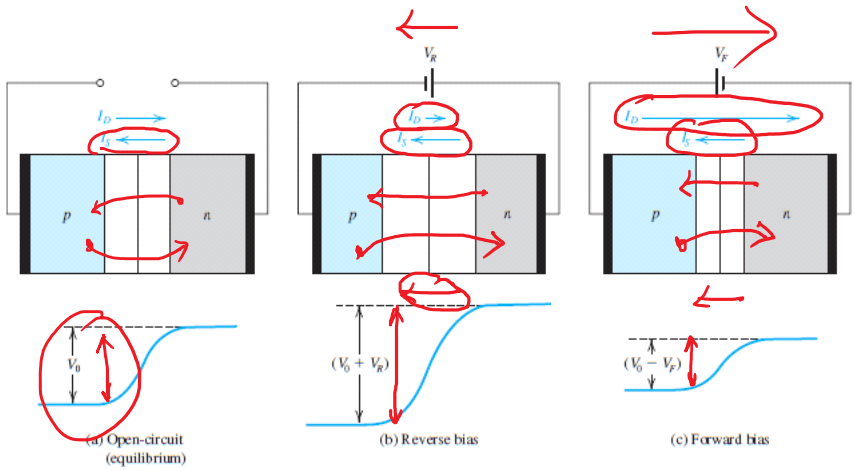
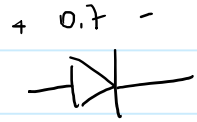
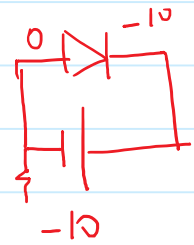
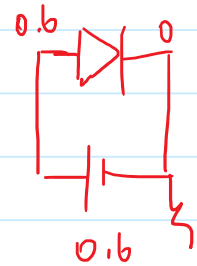


Figure 3.11 The *pn* junction in: (a) equilibrium; (b) reverse bias; (c) forward bias.







Analogy for Diffusion Current

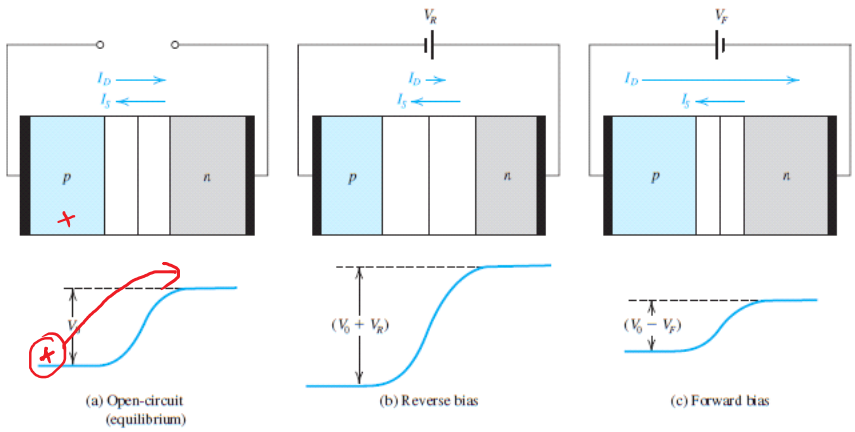
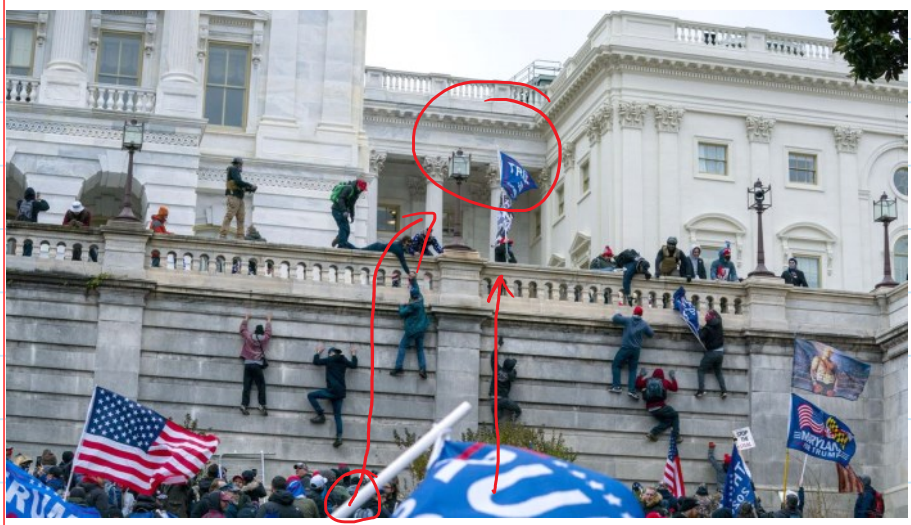
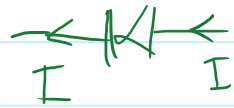
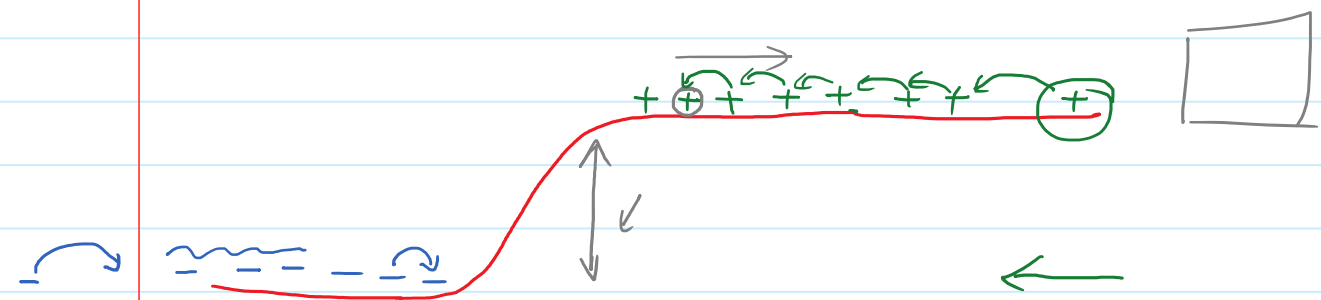
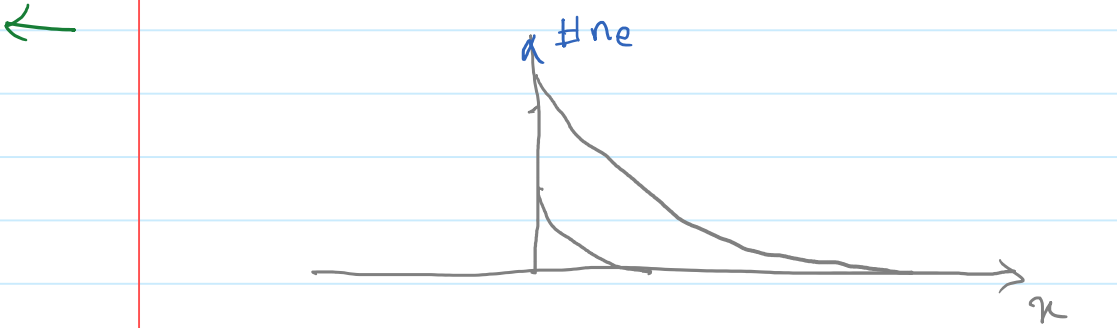


Figure 3.11 The *pn* junction in: (a) equilibrium; (b) reverse bias; (c) forward bias.

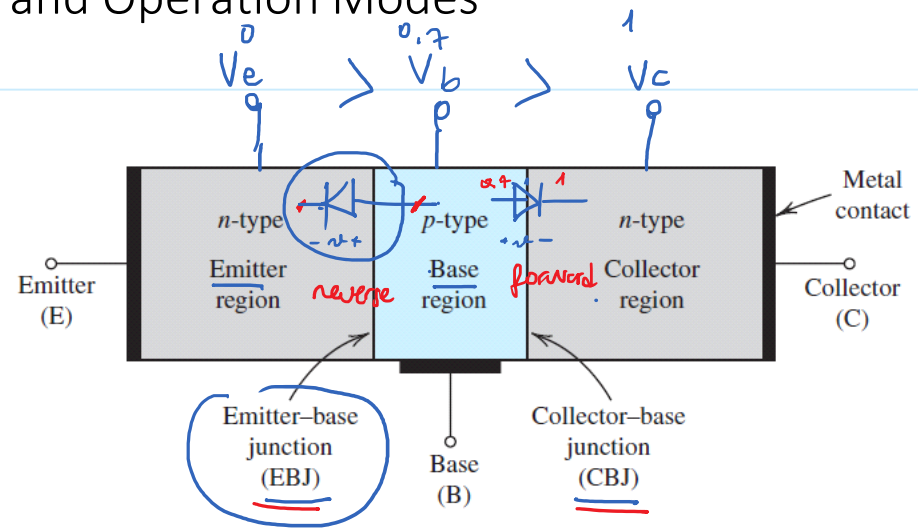








BJT and Operation Modes



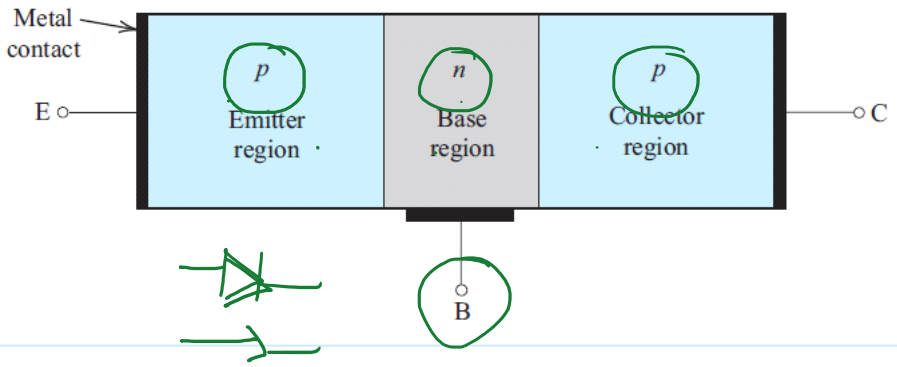
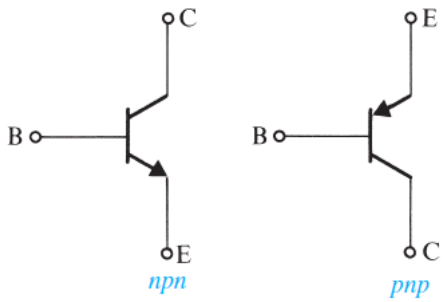
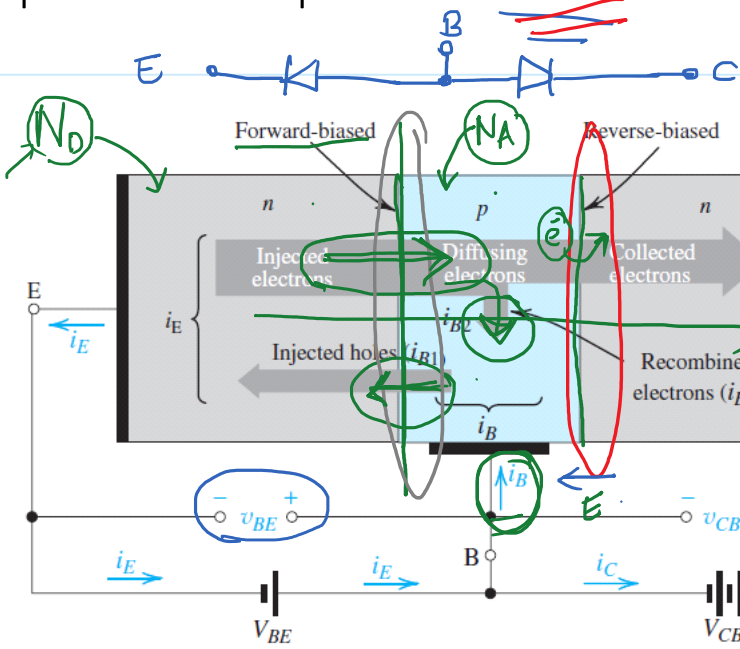


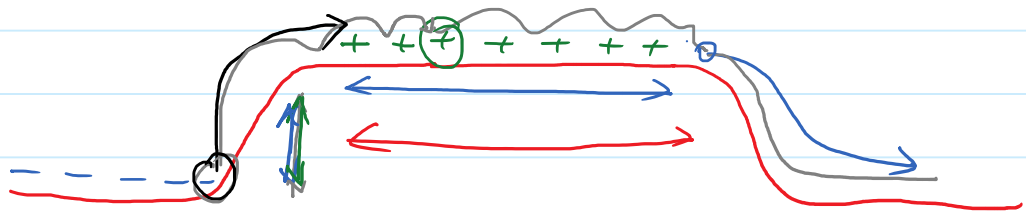
Table 6.1 BJT Modes of Operation

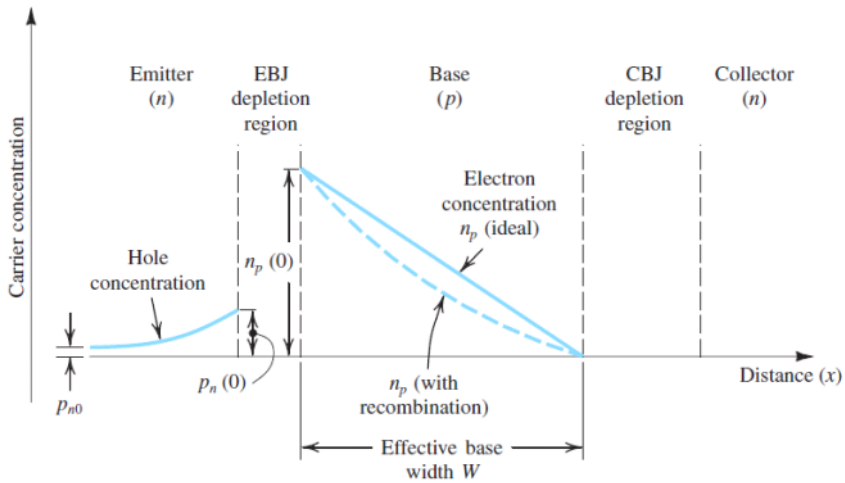
Mode	EBJ	CBJ
Cutoff	Reverse	Reverse
Active	Forward	Reverse
Saturation	Forward	Forward

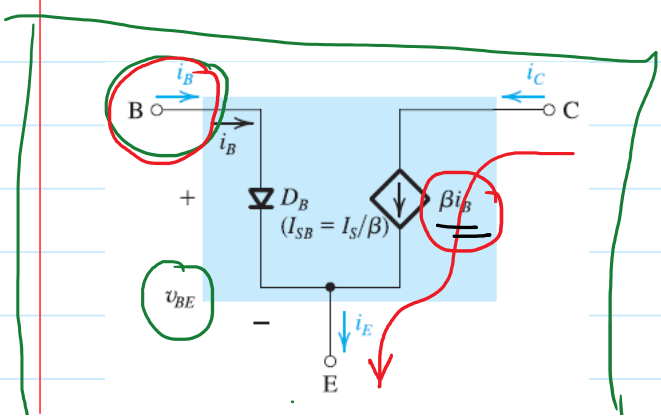


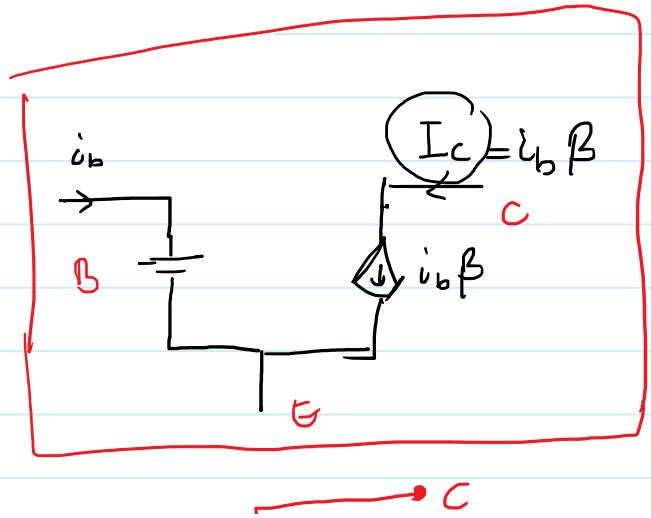
Operation of npn in the Active Mode

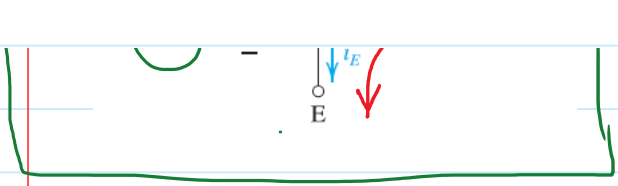


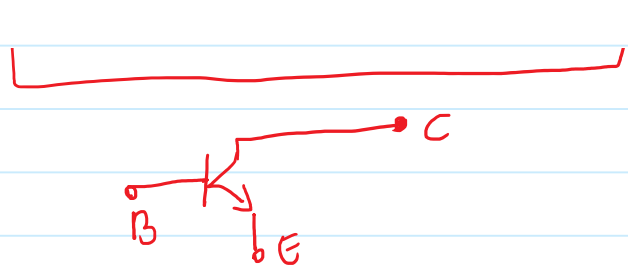




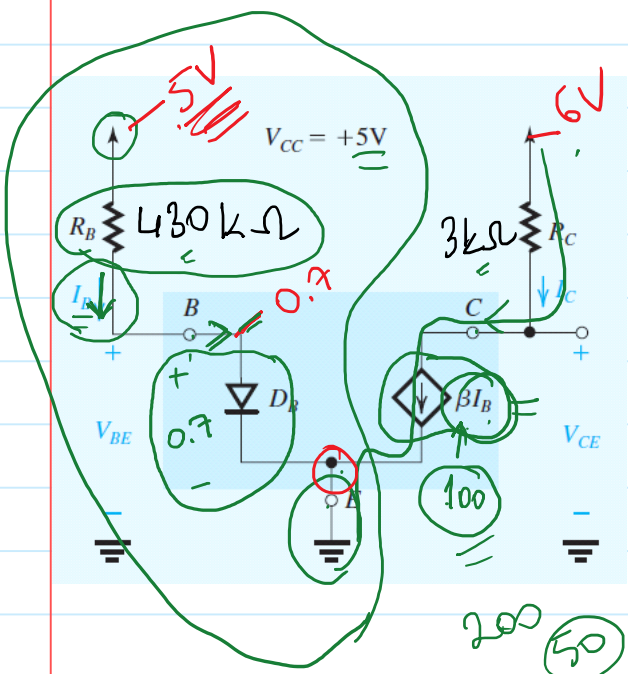





E
I_E


C
B
E

Example

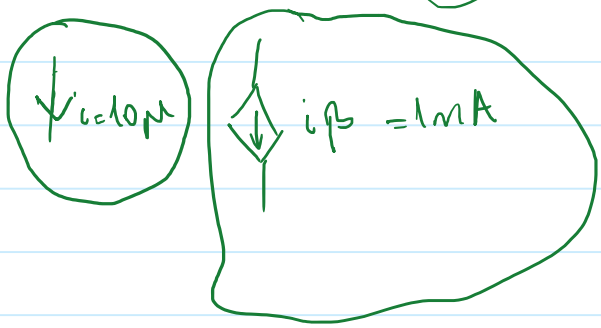


$= ?$ $V_{CE} = ?$

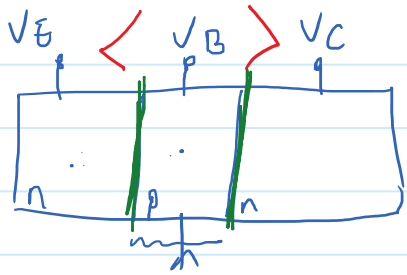
$$I_B = \frac{5 - 0.7}{430k} = \frac{4.3}{430k} = 10\mu A$$

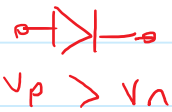
$10\,\mu A = 10 \times 10^{-6} A$

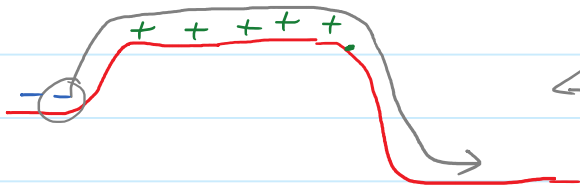
$I_C = 100 \cdot 10\mu A = 1mA$



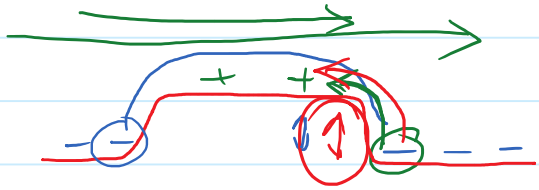
Operation of npn in the Saturation Mode



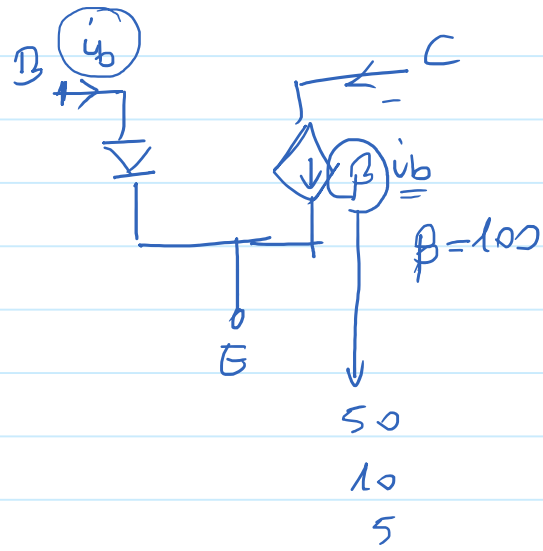




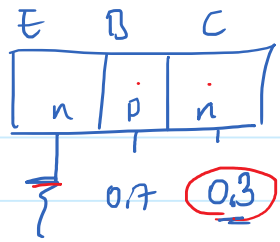
← active mode

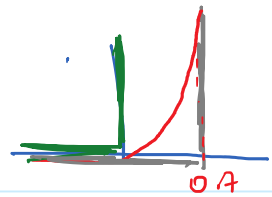


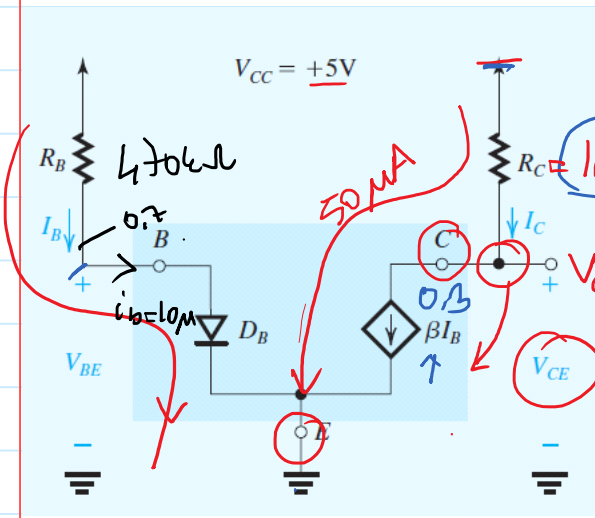




Template







$R_c = \frac{5 - 0,3}{0,1mA} = 47k\Omega$

$V_{CE} = V_C < 0,4$

$I_c = \frac{5 - 0,3}{100k} = \frac{4,7}{100k} = 50\mu A$

$\beta = 5$

$\beta_F = 100$

$\beta_S = 90$